DATA SCIENCE MINOR PROJECT REPORT

CRIME IN INDIA: A COMPREHENSIVE DATA ANALYSIS

Submitted by

Anubhaw Kumar Rai
Registration No.: 12405773
Section: 424ZC
Course Code: INT387

Under the Guidance of
Savleen Kaur Ma'am

Lovely School of Computer Science and Engineering
Lovely Professional University, Phagwara

CERTIFICATE

This is to certify that Anubhaw Kumar Rai bearing Registration No. 12405773 has completed the Data Science Minor Project titled, "CRIME IN INDIA: A COMPREHENSIVE DATA ANALYSIS" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor: _____

Designation of the Supervisor: Assistant Professor

School of Computer Science and Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 22 April 2026

DECLARATION

I, Anubhaw Kumar Rai, student of B.tech under CSE/IT Discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine. This report has been prepared as part of the Data Science Minor Project course requirements.

I further declare that this report has not been submitted previously to this institution or to any other institution for the award of any degree, diploma, or other qualification.

The analysis, interpretations, and conclusions stated in this report are entirely based on my own study and analysis of the dataset. I take full responsibility for the accuracy and integrity of the content presented herein.

Date: 22 April 2026 Signature: Anubhaw

Registration No.: 12405773 Name of the Student: Anubhaw Kumar Rai

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Anubhaw Kumar Rai
12405773

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1. INTRODUCTION

1.1 Background

Crime is a pervasive societal challenge that affects the safety, well-being, and development of communities across India. As one of the world's most populous nations with a diverse socio-economic landscape, India faces a complex and evolving crime scenario. The systematic collection, analysis, and visualization of crime data is essential for law enforcement agencies, policy makers, and researchers to understand crime patterns, allocate resources effectively, and design targeted preventive measures.

In recent years, the availability of digital crime records has opened unprecedented opportunities for data-driven analysis. Advanced data science techniques—including statistical analysis, data visualization, and exploratory data analysis (EDA)—enable us to extract actionable insights from large and complex crime datasets. This project leverages these methodologies to analyze a comprehensive crime dataset covering Indian cities from 2020 to 2024.

1.2 Problem Statement

Despite the large volume of crime data being recorded in India, there exists a significant gap in structured, accessible, and visually interpretable analysis of this data. Traditional reporting methods often fail to reveal underlying patterns, geographic disparities, temporal trends, and demographic vulnerabilities embedded in crime data. This project aims to bridge this gap by applying data science techniques to derive meaningful insights from a multi-year, multi-city crime dataset.

1.3 Dataset Overview

The dataset used in this project contains 40,160 crime records spanning the years 2020 to 2024, covering 20 major Indian cities including Delhi, Mumbai, Bangalore, Hyderabad, Kolkata, Chennai, and others. Each record includes 14 attributes capturing various dimensions of crime incidents:

Column Name	Data Type	Description
Report Number	Integer	Unique identifier for each crime report

Column Name	Data Type	Description
Date Reported	DateTime	Date when the crime was officially reported
Date of Occurrence	DateTime	Actual date when the crime occurred
Time of Occurrence	DateTime	Time when the crime took place
City	String	Indian city where the crime occurred
Crime Code	Integer	Numeric code assigned to the crime type
Crime Description	String	Descriptive category of the crime
Victim Age	Integer	Age of the crime victim (10-79 years)
Victim Gender	String	Gender of victim (M/F/X)
Weapon Used	String	Type of weapon involved
Crime Domain	String	Broad domain/category of crime
Police Deployed	Integer	Number of police officers deployed
Case Closed	String	Whether the case was resolved (Yes/No)
Date Case Closed	DateTime	Date when the case was officially closed

1.4 Significance of the Study

This analysis holds significant importance for multiple stakeholders. For law enforcement agencies, the findings can help in optimizing police deployment, identifying high-crime zones, and improving case resolution rates. For policymakers, the temporal and geographic analysis provides evidence-based insights for targeted crime prevention initiatives. For the academic community, the project demonstrates the application of data science methodologies in criminology, contributing to the growing field of computational social science.

The project also highlights the relationship between crime types, weapons used, victim demographics, and case closure rates—providing a holistic understanding of the crime landscape in urban India from 2020 to 2024.

2. OBJECTIVES / SCOPE OF THE ANALYSIS

2.1 Primary Objectives

The following primary objectives guide this data science project:

1. To perform a comprehensive Exploratory Data Analysis (EDA) on the crime dataset covering 20 major Indian cities from 2020 to 2024, uncovering key patterns and statistical distributions.
2. To analyze the geographic distribution of crimes across Indian cities and identify high-crime urban centers requiring enhanced law enforcement attention.
3. To examine temporal trends in crime occurrences across years and months, identifying seasonal patterns and year-on-year fluctuations.
4. To classify and analyze crime types by their description and domain, understanding the prevalence and distribution of various crime categories.
5. To study victim demographic profiles including age and gender distribution, identifying vulnerable population segments.
6. To evaluate weapon usage patterns across different crime types and cities, providing insights for law enforcement preparedness.
7. To assess case resolution rates and identify factors influencing the closure of criminal cases.
8. To analyze police deployment patterns relative to crime severity and domain.

2.2 Secondary Objectives

- To create meaningful data visualizations that communicate complex crime patterns in an accessible and interpretable manner.
- To perform cross-dimensional analysis examining relationships between variables such as crime type, city, victim demographics, and case outcomes.
- To establish a reproducible data analysis pipeline that can be applied to future crime datasets.
- To provide data-driven recommendations for crime prevention strategies and law enforcement resource allocation.

2.3 Scope of the Analysis

The scope of this analysis encompasses:

2.3.1 Geographic Scope

The analysis covers 20 major Indian cities: Delhi, Mumbai, Bangalore, Hyderabad, Kolkata, Chennai, Pune, Ahmedabad, Jaipur, Lucknow, Kanpur, Surat, Nagpur, Agra, Ludhiana, Bhopal, Patna, Varanasi, Meerut, and Visakhapatnam. These cities represent diverse geographic regions and urban population centers across India.

2.3.2 Temporal Scope

The dataset covers crime records from January 2020 through mid-2024 (approximately 4.5 years), enabling both annual trend analysis and intra-year seasonal pattern detection.

2.3.3 Crime Type Scope

The analysis covers 21 distinct crime categories including Burglary, Vandalism, Fraud, Domestic Violence, Firearm Offense, Robbery, Kidnapping, Identity Theft, Sexual Assault, Assault, Traffic Violation, Public Intoxication, Homicide, Cybercrime, Illegal Possession, Arson, Drug Offense, Extortion, Counterfeiting, Shoplifting, and Vehicle Theft.

2.3.4 Analytical Scope

The analytical scope includes descriptive statistics, frequency analysis, comparative analysis across cities and years, demographic analysis of victims, weapon usage patterns, and case resolution analysis. The project employs Microsoft Excel and Power BI for data processing, analysis, and visualization.

2.4 Limitations

- The dataset may not represent the complete universe of crimes, as some incidents may go unreported.
- The analysis is limited to reported crime data and does not account for crimes that were never registered with law enforcement.
- The dataset covers only selected major cities and may not represent rural or semi-urban crime patterns.

- Causal inferences cannot be drawn solely from statistical analysis; the findings represent correlational patterns.

3. SOURCE OF DATASET

3.1 Dataset Description

The primary dataset used in this project is the "Crime Dataset India" — a comprehensive compilation of crime incident records across major Indian metropolitan cities. The dataset was obtained from publicly available data repositories and contains structured information on crime incidents spanning from January 2020 to mid-2024.

3.2 Dataset Specifications

Parameter	Details
Dataset Name	Crime Dataset India (crime_dataset_india.xlsx)
File Format	Microsoft Excel Workbook (.xlsx)
Total Records	40,160 crime incidents
Number of Attributes	14 columns
Time Period	January 2020 – Mid 2024 (approx. 4.5 years)
Geographic Coverage	20 major Indian cities
Data Type	Structured, tabular data
Language	English
Missing Values	Present in Date Case Closed (for open cases)

3.3 Data Reliability and Quality

The dataset demonstrates several characteristics of high-quality data suitable for analytical purposes:

3.3.1 Completeness

With the exception of the 'Date Case Closed' field (which is legitimately empty for open cases), all other attributes are completely populated across all 40,160 records. This results in a data completeness rate of approximately 99.8% for actionable fields.

3.3.2 Consistency

The dataset maintains consistent data formats across all records. Date fields follow a uniform datetime format, categorical fields use standardized value sets, and numerical fields maintain consistent data types throughout the dataset.

3.3.3 Distribution

The crime records are distributed across 5 complete or near-complete years: 2020 (8,784 records), 2021 (8,760 records), 2022 (8,760 records), 2023 (8,760 records), and 2024 (5,096 records through approximately June 2024). The roughly uniform distribution across years ensures temporal comparability of the data.

3.4 Ethical Considerations

The dataset used in this project is synthetic or anonymized in nature, designed for educational and research purposes. No personally identifiable information (PII) of actual crime victims is present in the dataset. All analysis conducted in this project strictly adheres to ethical data science practices, including:

- Responsible use of sensitive crime-related data for academic purposes only
- No attempt to identify or profile specific individuals from the dataset
- Objective and unbiased interpretation of statistical findings
- Transparent reporting of methodology and limitations

3.5 Data Access

The dataset was processed and analyzed using Microsoft Excel for data transformation and basic statistical analysis, and Microsoft Power BI for advanced visualization and dashboard creation. The dataset file is provided as a supplementary attachment to this report in its original Excel format.

4. ETL PROCESS (EXTRACT, TRANSFORM, LOAD)

4.1 Overview of ETL

The Extract, Transform, and Load (ETL) process forms the foundational data pipeline for this project. It encompasses all data engineering activities performed to prepare the raw dataset for analysis. The ETL process ensures data quality, consistency, and analytical readiness before any statistical analysis or visualization is conducted.

4.2 Extract Phase

The extraction phase involved retrieving the raw crime dataset from its source format and loading it into the analytical environment.

4.2.1 Data Extraction Steps

9. Imported the crime_dataset_india.xlsx file into Microsoft Excel for initial inspection
10. Performed an initial data inventory to assess the dataset structure, dimensions, and attribute types
11. Identified 40,160 records with 14 attributes spanning the period 2020-2024
12. Exported relevant subsets of data to Power BI for visualization
13. Verified data integrity by checking record counts and attribute distributions

4.2.2 Initial Data Profile

Upon extraction, the following initial data profile was established:

- Total Records: 40,160
- Total Attributes: 14
- Date Range: January 1, 2020 to June 2024
- Cities Covered: 20 major Indian metropolitan cities
- Crime Categories: 21 distinct crime descriptions
- Null Values: Detected in 'Date Case Closed' field (for open/pending cases)

4.3 Transform Phase

The transformation phase is the most critical component of the ETL process, involving data cleaning, standardization, feature engineering, and quality assurance.

4.3.1 Data Type Standardization

All date fields (Date Reported, Date of Occurrence, Date Case Closed, Time of Occurrence) were validated and standardized to uniform datetime format (YYYY-MM-DD HH:MM:SS). Categorical fields were verified for consistent encoding without leading/trailing whitespace. Numeric fields were confirmed to contain only valid integer values within expected ranges.

4.3.2 Feature Engineering

New derived features were created to enable richer analysis:

14. Year: Extracted from 'Date of Occurrence' to enable year-wise trend analysis
15. Month: Extracted to enable seasonal pattern analysis
16. Days to Close: Calculated as (Date Case Closed - Date of Occurrence) for resolved cases — revealed a mean resolution time of 85 days
17. Case Closure Rate: Computed as the ratio of closed cases to total cases per category
18. Crime Rate per City: Normalized crime counts by city

4.3.3 Data Cleaning

The following data cleaning operations were performed:

- Verified that the 20,098 records with 'No' in Case Closed field correctly have null values in Date Case Closed — this is an expected and valid missing data pattern, not an error
- Removed any duplicate records (none found in this dataset)
- Validated victim age range: all values between 10 and 79 years — within acceptable bounds
- Validated police deployment range: 1 to 19 officers per incident
- Standardized gender codes: F (Female), M (Male), X (Non-binary/Unspecified)
- Verified crime code consistency with crime descriptions

4.3.4 Data Transformation Summary Table

Transformation	Action Taken	Result
Date Parsing	Converted to datetime format	100% success rate

Transformation	Action Taken	Result
Year Extraction	Derived from Date of Occurrence	5 year groups created
Null Handling	Marked open cases as 'Case Pending'	Handled 20,098 records
Feature Engineering	Computed Days to Close	85 days mean resolution
Data Validation	Range checks on numeric fields	All values within bounds
Categorical Check	Verified unique values per column	No inconsistencies found

4.4 Load Phase

Following transformation, the cleaned and enriched dataset was loaded into the analytical tools:

19. The Excel workbook with added derived columns was saved as the primary working dataset
20. The dataset was imported into Microsoft Power BI using direct Excel connection
21. Data relationships were established within Power BI's data model
22. Custom measures and DAX formulas were created in Power BI for advanced calculations
23. The processed dataset was verified by cross-checking aggregated statistics between Excel and Power BI

4.5 Tools Used in ETL

Tool	Purpose	Version
Microsoft Excel	Data extraction, transformation, statistical analysis	Microsoft 365
Microsoft Power BI	Data loading, modeling, visualization, dashboard creation	Desktop (2024)
Power Query Editor	Data transformation within Power BI	Built-in
DAX (Data Analysis Expressions)	Custom measures and calculations in Power BI	Built-in

5. ANALYSIS 1 – CRIME DISTRIBUTION BY CITY

5.1 Introduction

This analysis examines the geographic distribution of crimes across the 20 major Indian cities included in the dataset. Understanding which cities report the highest crime volumes is essential for prioritizing law enforcement resources and policy interventions. The analysis reveals significant disparities in crime rates across Indian cities, reflecting differences in population size, urbanization levels, reporting mechanisms, and socio-economic conditions.

5.2 General Description

The dataset covers 20 Indian cities with a combined total of 40,160 crime records over the period 2020-2024. Delhi emerges as the city with the highest absolute crime count, followed by Mumbai and Bangalore. The analysis employs frequency counts, percentage distributions, and per-city ranking to identify geographic crime hotspots.

5.3 Specific Requirements, Functions and Formulas

The following Excel functions and Power BI measures were used:

- COUNTIF(City Range, City Name): For counting crime incidents per city
- PERCENTAGE = City Count / Total Count * 100: For proportional distribution
- Power BI DAX: Count of Report Number = COUNT('crime_dataset_india'[Report Number])
- Power BI Map Visual: Geographic plotting using city names for location intelligence

5.4 Analysis Results

Rank	City	Crime Count	% of Total	Cumulative %
1	Delhi	5,400	13.44%	13.44%
2	Mumbai	4,415	10.99%	24.43%
3	Bangalore	3,588	8.93%	33.36%
4	Hyderabad	2,881	7.17%	40.53%
5	Kolkata	2,518	6.27%	46.80%

Rank	City	Crime Count	% of Total	Cumulative %
6	Chennai	2,493	6.21%	53.01%
7	Pune	2,212	5.51%	58.52%
8	Ahmedabad	1,817	4.52%	63.04%
9	Jaipur	1,479	3.68%	66.72%
10	Lucknow	1,456	3.62%	70.34%
11	Kanpur	1,112	2.77%	73.11%
12	Surat	1,111	2.77%	75.88%
13	Nagpur	1,053	2.62%	78.50%
14	Agra	764	1.90%	80.40%
15	Ludhiana	761	1.90%	82.30%

5.5 Key Findings

- Delhi accounts for 13.44% of all reported crimes, making it the highest crime-reporting city in the dataset. Its status as the national capital with over 20 million residents contributes to this high absolute volume.
- The top 3 cities (Delhi, Mumbai, Bangalore) together account for 33.36% of all crimes — approximately one-third of total crime volume concentrated in just 3 cities.
- The top 6 metropolitan cities account for over 53% of all reported crimes, indicating significant urban concentration of crime incidents.
- Delhi reports the most prevalent crimes as Firearm Offenses (291 cases), Cybercrime (286), and Fraud (281), suggesting a profile skewed toward modern and weapon-related crimes.
- Mumbai's top crimes include Sexual Assault (243), Illegal Possession (229), and Fraud (227), reflecting a different crime profile from Delhi.

5.6 Visualization

The geographic distribution was visualized using a Power BI Map visual (Bing Maps integration) plotting crime density by city using the 'City' field. A horizontal bar chart was also created showing 'Count of Report Number by City' sorted in descending order, clearly showing Delhi's dominance and the rapid drop-off as we move to smaller cities. The dashboard also shows a total report count of 5,643 visible in the top-right KPI card, representing the count visible for the current filter context.

6. ANALYSIS 2 – CRIME TRENDS OVER TIME (2020-2024)

6.1 Introduction

Temporal analysis of crime data reveals how crime patterns evolve over time, enabling identification of year-on-year trends and seasonal variations. This analysis examines crime occurrences across the 5-year period (2020-2024) and monthly patterns within years, providing insights into the dynamics of crime in Indian cities.

6.2 General Description

The dataset spans January 2020 through approximately June 2024. Year 2024 has fewer records (5,096) as the dataset does not cover the complete calendar year. The analysis employs time-series visualization, year-wise comparisons, and monthly distribution analysis.

6.3 Functions and Formulas

- YEAR(Date of Occurrence): Extracts year for annual grouping
- MONTH(Date of Occurrence): Extracts month for seasonal analysis
- COUNTIFS with date criteria for time-bounded counts
- Power BI: Time Intelligence functions and Date Hierarchy for drill-down capability

6.4 Year-wise Crime Distribution

Year	Crime Count	% of Total	Avg Monthly	Case Closure Rate
2020	8,784	21.87%	732	50.48%
2021	8,760	21.81%	730	49.39%
2022	8,760	21.81%	730	49.22%
2023	8,760	21.81%	730	50.94%
2024*	5,096	12.69%	849*	49.59%

* 2024 data covers approximately 6 months only

6.5 Monthly Crime Distribution

Month	Crime Count	% of Annual Avg
January	3,600	103%
February	3,288	94%
March	3,600	103%
April	3,480	100%
May	3,600	103%
June	3,480	100%
July	3,584	103%
August	3,144	90%
September	3,048	87%
October	3,144	90%
November	3,048	87%
December	3,144	90%

6.6 Key Findings

- Crime volume remained remarkably stable from 2020 to 2023, with approximately 8,760 incidents per year (averaging 730/month), suggesting a consistent systemic crime pattern rather than dramatic fluctuations.
- The years 2020-2023 show near-identical crime counts (within 0.3%), indicating that the dataset has a consistent reporting density across these years.
- Monthly analysis reveals a slight seasonal pattern: January, March, May, and July show higher crime counts (3,580-3,600), while August, September, November, and December show lower counts (3,048-3,144).
- Case closure rates have remained relatively stable between 49-51% across all years, indicating consistent law enforcement performance.
- The Power BI area chart visualization of 'Count of Case Closed by Year' shows consistent performance trends, with the visualization revealing the purple shaded area representing the volume of cases across the years 2020-2024.

7. ANALYSIS 3 – CRIME CATEGORY AND DOMAIN ANALYSIS

7.1 Introduction

Understanding the distribution of crime types is fundamental to crime prevention strategy. This analysis examines the prevalence of 21 distinct crime categories and 4 broad crime domains, identifying which types of crimes are most common in Indian cities and how they are distributed across the crime landscape.

7.2 Crime Domain Distribution

Crime Domain	Count	Percentage	Description
Other Crime	22,948	57.14%	Non-violent, non-traffic general crimes
Violent Crime	11,472	28.56%	Crimes involving physical violence or threat
Fire Accident	3,825	9.52%	Arson and fire-related incidents
Traffic Fatality	1,915	4.77%	Traffic-related crimes and fatalities

7.3 Top Crime Descriptions

Rank	Crime Description	Count	% of Total	Cases Closed
1	Burglary	1,980	4.93%	954
2	Vandalism	1,975	4.92%	965
3	Fraud	1,965	4.89%	984
4	Domestic Violence	1,932	4.81%	985
5	Firearm Offense	1,931	4.81%	959
6	Robbery	1,928	4.80%	961
7	Kidnapping	1,920	4.78%	956
8	Identity Theft	1,918	4.78%	965
9	Sexual Assault	1,917	4.77%	929
10	Assault	1,915	4.77%	952
11	Traffic Violation	1,915	4.77%	960

Rank	Crime Description	Count	% of Total	Cases Closed
12	Public Intoxication	1,912	4.76%	1,000
13	Homicide	1,909	4.75%	950
14	Cybercrime	1,899	4.73%	945
15	Illegal Possession	1,895	4.72%	959

7.4 Key Findings

- The 21 crime types are remarkably evenly distributed, each accounting for approximately 4.7-4.9% of total crimes, suggesting the dataset represents a balanced sample across all crime categories.
- Other Crime constitutes the majority domain at 57.14%, encompassing property crimes, financial crimes, and social offenses that are non-violent in nature.
- Violent Crime at 28.56% represents a significant proportion of criminal activity, encompassing assault, homicide, kidnapping, sexual assault, and domestic violence.
- Public Intoxication has the highest case closure rate (1,000 of ~1,912 cases = 52.3%), likely due to the straightforward nature of evidence gathering.
- Sexual Assault shows the lowest case closure rate (929/1,917 = 48.5%), reflecting the well-documented challenges in prosecuting such cases.
- The balanced distribution across crime types makes this dataset suitable for comparative analysis without the bias that would come from highly skewed distributions.

8. ANALYSIS 4 – VICTIM DEMOGRAPHICS ANALYSIS

8.1 Introduction

Victim demographics analysis provides crucial insights into which population segments are most vulnerable to crime. This analysis examines the age distribution and gender composition of crime victims in the dataset, helping identify at-risk groups and informing targeted victim support and crime prevention programs.

8.2 Victim Age Analysis

The victim age distribution spans from 10 to 79 years, providing a comprehensive view of crime victimization across age groups.

8.2.1 Age Distribution Statistics

Statistic	Value
Total Victims	40,160
Minimum Age	10 years
Maximum Age	79 years
Mean Age	44.49 years
Median Age	44 years
Standard Deviation	20.23 years
25th Percentile (Q1)	27 years
75th Percentile (Q3)	62 years
Interquartile Range	35 years

8.2.2 Age Group Analysis

Age Group	Range	Characteristics
Youth	10-20 years	Young victims, school-age group
Young Adult	21-30 years	College-age, early career group
Adult	31-50 years	Prime working age adults
Middle-Aged	51-65 years	Established working adults
Senior	66-79 years	Elderly victims, potentially vulnerable

The near-uniform distribution of victim ages (mean 44.49, median 44) with a standard deviation of 20.23 years indicates that crime victimization affects individuals broadly across all age groups. The donut chart visualization in the Power BI dashboard ('Sum of Victim Age by Year') breaks down victim age across the 5 years, showing year-wise distribution with 2020 leading at approximately 31K in total victim age sum.

8.3 Victim Gender Analysis

Gender	Code	Count	Percentage	Avg Age
Female	F	22,423	55.83%	44.27 years
Male	M	13,405	33.38%	44.54 years
Non-binary/Other	X	4,332	10.78%	45.48 years

8.4 Key Findings

- Female victims constitute 55.83% of total crime victims — the majority — indicating that women face significantly higher victimization risk across the covered crime types, which include domestic violence, sexual assault, and identity theft.
- Male victims at 33.38% are the second largest group, primarily affected by physical altercations, robbery, and drug-related offenses.
- Non-binary/Other gender individuals represent 10.78% of victims, a notably high proportion suggesting the dataset captures gender-diverse victim reporting.
- Average victim age is remarkably similar across all gender groups (44.27 to 45.48 years), suggesting that age vulnerability is relatively uniform across gender categories.
- The scatter plot in the Power BI dashboard ('Count of Victim Gender by Victim Age') shows the distribution of gender across the 10-79 age range, with the wave-like pattern indicating relatively uniform victimization across all ages with slight peaks in the 20-40 range.
- The near-equal median ages across genders suggest that middle age (40s) represents a period of elevated crime victimization across all gender categories.

9. ANALYSIS 5 – WEAPON USAGE ANALYSIS

9.1 Introduction

Weapon usage in crime incidents provides critical intelligence for law enforcement preparedness, policy development, and preventive interventions. This analysis examines the distribution of weapons used across crime incidents in the dataset, their relationship with crime severity, and geographic patterns of weapon prevalence.

9.2 Overall Weapon Distribution

Weapon Type	Count	Percentage	Case Closure Rate
Knife	5,835	14.53%	49.8%
Explosives	5,751	14.32%	49.7%
Blunt Object	5,737	14.28%	50.1%
Poison	5,728	14.26%	49.6%
Other	5,676	14.13%	50.2%
Firearm	5,643	14.05%	49.7%

9.3 General Description

The dataset captures 6 distinct weapon types used in crime incidents: Knives, Explosives, Blunt Objects, Poison, Other (unspecified weapons), and Firearms. Notably, the Power BI dashboard displays the KPI count as 5,643 — corresponding to the count of Firearm offenses when the Firearm filter is applied, consistent with the dashboard screenshot showing 'Weapon Used' filter panel.

9.4 Functions and Formulas Used

- COUNTIF(Weapon Column, Weapon Type): Count of each weapon type
- Power BI Slicer: Weapon Used filter for interactive cross-filtering of all visuals
- Case Closure Rate = DIVIDE(COUNTIF(Case Closed, 'Yes'), COUNT(Case Closed))
- DAX Measure: % Weapon Use = DIVIDE([Weapon Count], [Total Crimes], 0)

9.5 Key Findings

- All six weapon types are strikingly evenly distributed (14.05% to 14.53%), each accounting for approximately one-sixth of all incidents. This near-perfect balance suggests systematic data collection across all weapon types.
- Knives are the most frequently used weapon (5,835 incidents, 14.53%), followed closely by Explosives (5,751 incidents, 14.32%).
- The case closure rate is remarkably consistent across all weapon types (49.6% to 50.2%), suggesting that the type of weapon used does not significantly influence case resolution probability.
- The presence of Explosives as the second most common weapon type (5,751 cases) is notable and likely encompasses a range of incendiary devices rather than conventional military explosives.
- Firearms, while ranking last in frequency at 14.05%, remain a significant concern, accounting for 5,643 incidents across the 5-year period.
- The even distribution across weapon types challenges the common assumption that certain weapons (e.g., firearms or knives) dominate crime incidents, suggesting diverse criminal methodologies in Indian cities.

The Power BI dashboard's filter panel (right side) shows the Weapon Used slicer with checkboxes for Blunt Object, Explosives, Firearm, Knife, None, Other, and Poison — enabling interactive drill-down by weapon type across all dashboard visuals.

10. ANALYSIS 6 – CASE RESOLUTION ANALYSIS

10.1 Introduction

Case resolution analysis evaluates the effectiveness of law enforcement in closing criminal cases. The ability to close cases — through arrests, prosecutions, or administrative resolution — is a key performance indicator for police departments. This analysis examines overall resolution rates, trends over time, and variations across crime types.

10.2 Overall Case Resolution

Status	Count	Percentage
Cases Closed (Resolved)	20,062	49.95%
Cases Open (Pending)	20,098	50.05%
Total Cases	40,160	100%

10.3 Year-wise Case Resolution Rates

Year	Total Cases	Closed Cases	Open Cases	Closure Rate
2020	8,784	4,434	4,350	50.48%
2021	8,760	4,327	4,433	49.39%
2022	8,760	4,312	4,448	49.22%
2023	8,760	4,462	4,298	50.94%
2024*	5,096	2,527	2,569	49.59%

10.4 Case Resolution by Crime Category

Crime Type	Closed Cases	Total Cases	Closure Rate
Public Intoxication	1,000	1,912	52.3%
Domestic Violence	985	1,932	51.0%
Fraud	984	1,965	50.1%
Identity Theft	965	1,918	50.3%
Vandalism	965	1,975	48.9%

Crime Type	Closed Cases	Total Cases	Closure Rate
Robbery	961	1,928	49.8%
Traffic Violation	960	1,915	50.1%
Homicide	950	1,909	49.7%
Shoplifting	950	~1,900	50.0%
Sexual Assault	929	1,917	48.5%

10.5 Average Time to Close Cases

Metric	Value
Mean Days to Close	85.1 days
Median Days to Close	54 days
Standard Deviation	157.6 days
25th Percentile	19 days
75th Percentile	97 days
Maximum Days to Close	1,028 days

10.6 Key Findings

- The overall case closure rate of approximately 49.95% indicates that roughly half of all reported crimes in India's major cities remain unresolved — a significant challenge for the justice system.
- Case closure rates have remained stable across all 5 years (range: 49.22% to 50.94%), suggesting systemic factors rather than year-specific events drive resolution rates.
- Public Intoxication has the highest closure rate (52.3%), likely because such cases involve direct observation and straightforward evidence collection.
- Sexual Assault cases have the lowest closure rate (48.5%), consistent with global trends where evidentiary challenges, underreporting, and systemic barriers reduce prosecution success.
- The median resolution time of 54 days is substantially lower than the mean (85 days), indicating that the distribution of case resolution times is right-skewed — a small number of complex cases take very long to resolve, pulling the mean upward.
- Some cases took over 1,000 days to close, suggesting the presence of complex, multi-year investigations in the dataset.

- The Power BI area chart visualization 'Count of Case Closed by Year' shows the consistent case volume with the purple shaded area representing year-wise case activity.

11. ANALYSIS 7 – POLICE DEPLOYMENT ANALYSIS

11.1 Introduction

Police deployment — the number of officers assigned to each crime incident — reflects law enforcement capacity, prioritization, and resource allocation strategies. This analysis examines how police deployment varies across crime domains, cities, and whether deployment levels correlate with case resolution outcomes.

11.2 Overall Police Deployment Statistics

Statistic	Value
Mean Officers Deployed	10.01 per incident
Median Officers Deployed	10 per incident
Minimum Deployment	1 officer
Maximum Deployment	19 officers
Standard Deviation	5.47 officers
25th Percentile (Q1)	5 officers
75th Percentile (Q3)	15 officers

11.3 Police Deployment by Crime Domain

Crime Domain	Avg Officers Deployed	Total Incidents	Total Officers Deployed
Traffic Fatality	10.04	1,915	~19,227
Other Crime	10.02	22,948	~229,919
Violent Crime	9.99	11,472	~114,605
Fire Accident	9.97	3,825	~38,136

11.4 Functions and Formulas Used

- AVERAGE(Police Deployed): Mean deployment calculation
- AVERAGEIF(Crime Domain, Domain Name, Police Deployed): Domain-specific deployment averages

- CORREL(Police Deployed, Case Closed Numeric): Correlation analysis
- Power BI DAX: Average of Police Deployed = AVERAGE('crime_dataset_india'[Police Deployed])
- Scatter plot: 'Count of Report Number and Average of Victim Age by City' in Power BI showing deployment distribution by city

11.5 Key Findings

- The mean police deployment of 10.01 officers per incident (median: 10) suggests a uniform deployment policy, with the dataset showing near-perfectly balanced deployment across all incident types.
- The remarkably similar average deployment across all 4 crime domains (9.97 to 10.04 officers) suggests that the current dataset records a standardized deployment value rather than capturing actual resource-differentiated deployment decisions.
- The wide range from 1 to 19 officers indicates that while the average is stable, individual incidents receive substantially different deployment levels based on severity assessment.
- The Power BI scatter plot ('Count of Report Number and Average of Victim Age by City') in the dashboard visualizes city-level deployment patterns alongside victim age averages, allowing multi-dimensional comparison.
- Optimizing police deployment based on crime type, severity, and geographic factors could potentially improve case closure rates, particularly for the 50% of cases currently remaining unresolved.
- Resource allocation modeling using machine learning approaches (e.g., predictive deployment models) could be a valuable next step building on the findings of this analysis.

11.6 Recommendations for Police Deployment

24. Implement risk-based deployment models that allocate more officers to high-severity crimes (Violent Crime domain) rather than maintaining uniform deployment across all crime types.
25. Prioritize enhanced deployment in Delhi, Mumbai, and Bangalore — the top three crime-volume cities — to address the concentrated crime load in these urban centers.

26. Develop specialized rapid-response teams for time-sensitive crimes like Firearm Offenses and Kidnapping, where quick deployment can significantly impact case outcomes.
27. Leverage temporal analysis findings to increase patrol presence during peak crime months (January, March, May, July) as a proactive crime prevention measure.

12. LIST OF ANALYSES WITH RESULTS

12.1 Summary of All Analyses Conducted

The following table provides a consolidated summary of all seven analyses conducted in this project, along with their primary findings and key insights:

Analysis	Focus Area	Method	Primary Result
1	Geographic Distribution	City-wise frequency analysis, Map visualization	Delhi (13.44%) is highest crime city; Top 3 cities = 33% of all crimes
2	Temporal Trends	Year-wise & monthly trend analysis	Stable ~8,760 crimes/year (2020-2023); Jan/Mar/May highest monthly counts
3	Crime Categories	Frequency distribution, domain classification	21 balanced crime types; Other Crime (57%) is dominant domain
4	Victim Demographics	Age & gender statistical analysis	55.83% female victims; Mean age 44.49 years; uniform across genders
5	Weapon Usage	Weapon frequency & closure rate analysis	6 weapons near-equally used; Knife most common (14.53%); ~50% closure across all
6	Case Resolution	Closure rate, time-to-close analysis	49.95% overall closure; Median 54 days; Sexual Assault lowest (48.5%)
7	Police Deployment	Deployment statistics & domain analysis	Mean 10 officers/incident; Uniform across domains; Range 1-19 officers

12.2 Cross-Analysis Insights

Examining the results across all seven analyses reveals several important cross-dimensional insights:

12.2.1 Geographic-Temporal Nexus

The combination of geographic analysis (Analysis 1) and temporal analysis (Analysis 2) reveals that the high crime volume in Delhi, Mumbai, and Bangalore is consistent across all years studied. These cities do not show temporary spikes but rather sustained elevated crime reporting throughout the 2020-2024 period.

12.2.2 Crime Type and Resolution Relationship

Comparing crime categories (Analysis 3) with case resolution (Analysis 6) shows that crimes with direct physical evidence (Public Intoxication, Domestic Violence) have higher closure rates, while crimes requiring forensic investigation or victim testimony (Sexual Assault, Cybercrime) face lower resolution rates. This pattern highlights the need for enhanced forensic capacity.

12.2.3 Weapon Use and Crime Severity

The near-uniform distribution of weapon types (Analysis 5) combined with the consistent case closure rates across all weapon types suggests that weapon type alone is not the primary determinant of case resolution. Other factors — such as witness availability, evidence quality, and crime reporting timeliness — likely play more significant roles.

12.2.4 Demographic Vulnerability Patterns

The victim demographics analysis (Analysis 4) combined with crime type analysis (Analysis 3) suggests specific vulnerability patterns: female victims are disproportionately represented in the dataset, with crimes like Domestic Violence and Sexual Assault being particularly impactful on female victims across all age groups.

12.3 Dashboard Observations

The Power BI dashboard created for this project (as shown in the screenshot provided) integrates all seven analyses into an interactive, multi-visual display. The dashboard features:

- KPI Card: Total count of 5,643 reports displayed prominently (filtered view)
- Horizontal Bar Chart: Count of Time of Occurrence by Year (2020-2024) showing annual distribution
- Area Chart: Count of Case Closed by Year showing temporal resolution patterns
- Donut Chart: Sum of Victim Age by Year with year-wise color coding (2020-2024)
- Geographic Map: Crime distribution mapped across Indian cities using Bing Maps
- Line Chart: Count of Victim Gender by Victim Age showing demographic distribution

- Scatter Plot: Count of Report Number and Average Victim Age by City for city-level analysis
- Weapon Used Slicer: Interactive filter panel for weapon-type cross-filtering
- Q&A Feature: Natural language query interface for ad-hoc data exploration

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14.3 Online Resources and Datasets

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- Microsoft Power BI Documentation. https://docs.microsoft.com/power-bi (accessed 2025)
- Towards Data Science – Medium Blog (Various Articles on Crime Data Analysis, 2022-2024)
- Government of India Open Data Platform. https://data.gov.in (accessed 2025)

14.4 Software and Tools

- Microsoft Excel (Microsoft 365) – Data processing, ETL, and statistical analysis
- Microsoft Power BI Desktop (2024 version) – Dashboard creation and data visualization
- Power Query Editor (built into Power BI) – Data transformation and cleaning
- DAX Studio – DAX formula testing and optimization

— END OF REPORT —

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DRIVE LINK : https://drive.google.com/file/d/1FaK6KqncNaSfwiWbVmUWu4e-Y-QRrslG/view?usp=sharing